

lec 1.4

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DBSCAN

1. What is DBSCAN?

DBSCAN (Density-Based Spatial Clustering of Applications with Noise) is an **Unsupervised Machine Learning Algorithm** that groups together data points that are **closely packed (high-density regions)** and identifies isolated points as **noise (outliers)**.

Unlike K-Means:

- DBSCAN **does not require the number of clusters (K)** before training.
- It can discover **clusters of arbitrary (irregular) shapes**.
- It can automatically detect **noise/outliers**.

Full Form

DBSCAN
=
Density-Based Spatial Clustering of
Applications with Noise

Main Idea

1. Dataset
2. Find Dense Regions
3. Expand Clusters
4. Separate Sparse Regions
5. Detect Noise Points

2. Why was DBSCAN Developed?

Traditional clustering algorithms like K-Means have several limitations.

- ✗ Must specify K beforehand
- ✗ Cannot detect noise
- ✗ Works best for spherical clusters
- ✗ Sensitive to outliers

DBSCAN solves these problems by using **density** instead of centroids.

